

Minh Phuc Nguyen

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EDUCATION

Union College

Schenectady, NY

Bachelor of Science in Computer Engineering

Expected graduation: June 2027

- Relevant Courses: Object-Oriented Programming, Data Structures & Algorithms, Parallel and High Performance Computing, CUDA Advanced Libraries, Computer Architecture, Embedded Systems, Machine Learning

EXPERIENCE

Research Assistant - Matrix Multiplication Project

Schenectady, NY

Union College's Computer Lab, Advisor: Professor Matthew Anderson

01/2024 - 02/2026

- Accelerated searches for SUSP tensors—algebraic structures that support faster matrix multiplication—with a packed, batched CUDA pipeline and asynchronous data transfers.
- Ran large-scale searches across two NVIDIA GeForce RTX 5070 Ti GPUs using device-aware scheduling, achieving up to $5.75\times$ higher end-to-end throughput with one GPU and $11.58\times$ with two GPUs compared with the original search implementation.
- Used Nsight Systems to profile the CPU-GPU pipeline and identify bottlenecks, increasing device utilization from 35% to consistently above 90%.

Research Assistant - Neural Receiver Project

Schenectady, NY

Union College ECE Department, Advisor: Professor Umair Mohammad

03/2026 - Present

- Built a 1.02M-parameter neural receiver in PyTorch for a coded 2×2 MIMO-OFDM link, replacing LMMSE equalization and APP demapping with a residual CNN while retaining LS channel estimation and LDPC decoding.
- Benchmarked the frozen receiver across 45 3GPP TDL environments, 11 SNR levels, and three evaluation seeds, processing more than 10 million coded bits per seed.
- Reduced mean post-LDPC Bit Error Rate from 0.04938 to 0.04144 at 16 dB relative to the classical LS/LMMSE/APP receiver, achieving lower mean BER in all 45 environments.

Undergraduate Thesis - Autonomous Driving Semantic Segmentation

Schenectady, NY

Union College, Advisor: Professor Cherrice Traver

03/2026 - Present

- Fine-tuned a SegFormer model for drivable-area segmentation and deployed it through ONNX and TensorRT using persistent execution contexts, reusable CUDA buffers, fused kernels, and an NVDEC/VIC video pipeline.
- Processed a complete 3,749-frame 1080p evaluation video in 88.84 seconds, achieving 42.20 FPS end to end across decoding, preprocessing, inference, visualization, and encoding.

Undergraduate Researcher - Approximate Nearest Neighbor Search

Schenectady, NY

Union College

01/2026 - 06/2026

- Built a two-stage multi-GPU vector-search pipeline using NVIDIA RAPIDS cuVS and a custom C++/CUDA exact candidate-reranking module.
- Benchmarked IVF-PQ and compressed CAGRA on five million 1,536-dimensional embeddings across two NVIDIA GeForce RTX 5070 Ti GPUs; achieved 158.2k QPS at 0.9471 Recall@10—63.4% higher throughput than IVF-PQ at comparable recall.

GPU Software Engineer

Undisclosed Hedge Fund

- Built a C++/CUDA simulation engine for financial risk and pricing workloads, adding CPU-GPU correctness checks to validate numerical results.
- Optimized GPU market-data processing with CUDA streams and OpenMP; used Nsight Systems to overlap computation and memory transfers, achieving a 10x speedup over the baseline kernels.
- Implemented GPU numerical routines with CUB and Thrust, including reductions and prefix scans used in quantitative-modeling pipelines.

PROJECTS

Autonomous Driving Object Detection System

Schenectady, NY

Embedded Vision Project

01/2025 - 06/2025

- Trained an object detection model on $2\times$ GPUs with data parallelism, overlapping gradient synchronization with backpropagation to reduce training time by 60% versus a single-GPU run.
- Optimized Jetson Orin Nano inference with TensorRT by converting the model to a static graph and fusing conv, bias, and ReLU operations, reaching 20 FPS at 1080p.

GPU-Accelerated OFDM-IM Simulation

Schenectady, NY

GPU and Communications Research Project

- Built CPU and persistent-kernel CUDA simulators for orthogonal frequency-division multiplexing with index modulation (OFDM-IM).
- Used CUB reductions and cuRAND generation to keep Monte Carlo trials on the GPU and reduce kernel-launch and data-transfer overhead.
- Added reproducible BER, runtime, GFLOPS, and resource-usage tooling; achieved a $474.5\times$ median speedup at the largest recorded workload, reducing runtime from 19.68 seconds on CPU to 41.47 milliseconds on GPU.

ktorch: CUDA Tensor Runtime

Schenectady, NY

In-Progress Private Project

- Built a CUDA-first C++ tensor runtime with pybind11 bindings, implementing F32 linear, layer-normalization, ReLU, softmax, in-place SGD, tensor views, cloning, host-device copies, and multi-stream execution.
- Benchmarked F32 linear-forward operations against PyTorch across 10 MLP and transformer matrix shapes on an NVIDIA GeForce RTX 5070 Ti, achieving a $1.18\times$ median speedup, $1.77\times$ peak speedup, and 26.4 TFLOP/s.

CUDA-based Real-Time Video Stream Filtering

Schenectady, NY

Personal Project

- Built a benchmark system to compare GPU filtering kernels using CUB, CUDA streams, shared memory tiling, and other CUDA optimizations; the CUB version achieved 52% higher FPS.
- Improved video-processing throughput by 70%+ using pinned host memory, multi-stream execution, warp-level primitives, and Thrust.

RESEARCH MANUSCRIPTS & PRESENTATIONS

- GPU Pipeline for Accelerated Search of an Upper Bound on the Matrix Multiplication Exponent. Unpublished manuscript, 2026.
- Batched GPU Execution for High-Throughput Combinatorial Search. Submitted to the SC26 ACM Student Research Competition; decision pending, 2026.
- Full-Grid Neural Reception for Coded 2×2 MIMO-OFDM Across 3GPP TDL Channels. Unpublished manuscript, 2026.

PROFESSIONAL TRAINING

GPU Programming Specialization (4 courses)

Completed

Johns Hopkins University, Coursera

Introduction to High-Performance and Parallel Computing

Completed

University of Colorado Boulder, Coursera

TECHNICAL SKILLS

Programming Language: C++, CUDA, Python, Java, Bash, Makefile

GPU & Parallel Computing: Nsight Systems, CUDA Streams, OpenMP, CUB, Thrust, TensorRT

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas

Tools & Platforms: Git, Valgrind, Linux Terminal, CMake